

Boyang (Boe) Zhou

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Education

University of Washington, Seattle Anticipated 6/2026
M.S. in Computer Science, GPA 3.9/4.0

University of Washington, Seattle 9/2021 – 3/2025
B.S. in Computer Science, Minor in Environmental Studies, Mary Gates Scholar, GPA 3.96/4.0

Research experience

Student Researcher - HCI, Computer Vision 11/2024 – Present
Makeability Lab, University of Washington, Seattle, WA
Advised by Prof. Jon E. Froehlich

- Leading a 3-person research team on exploring the design space of ability-based personal sensing. Developed a multi-model sensing pipeline with a privacy-preserving design for mobile devices, enabling always-on obstacle sensing to assist users in complex urban environments through visual, tactile, and audio feedback. The work is on track to be submitted to UIST 2026.
- Designed formative user study that involves both interview and field testing parts, clearly articulating both qualitative and quantitative metrics. Incorporated novel behavior data such as real-time gaze directions.
- Fine-tuned open-vocabulary visual segmentation models on a SoccerNet dataset and built a serialized inference pipeline to run locally on the Quest VR headset. This achieved on-device inference to power a visual augmentation system for blind and low-vision people to experience live soccer events.

Student Researcher - 3D/4D Reconstruction 6/2022 – Present
Reality Lab, University of Washington, Seattle, WA
Advised by Prof. Steve Seitz and Prof. Ira Kemelmacher-Shlizerman

- Contributing to 4D Gaussian Splat project. Employing prompt-to-prompt flow-matching method to remove reconstruction artifacts. The work is on track to be submitted to SIGGRAPH 2026.
- Developed a novel joint fine-tuning pipeline for the attention layers of video diffusion models to enable temporally consistent frame interpolation in animation workflows, reducing training cost by fine-tuning only 2% of model parameters while preserving motion expressiveness.
- Curated motion-rich video dataset and orchestrated large-scale training experiments across HPC GPU clusters to evaluate model performance under varying dataset and architectural configurations.
- Developed a pipeline for Finite Element Analysis simulation based on lumbar models reconstructed from MRI scans. The model has been used in an AR-based lumbar injection training simulator using Microsoft HoloLens, enabling a prototype for a radiation-free Epidural Steroid Injection procedure presented at the Annual Meeting of the American Society of Pain and Neuroscience.

Publications

Xiaojuan Wang, Boyang Zhou, Brian Curless, Ira Kemelmacher, Aleksander Holynski, Steve Seitz. 2025. **Generative Inbetweening: Adapting Image-to-Video Models for Keyframe Interpolation**. In *Proceedings of the International Conference on Learning Representations (ICLR 2025)*.

Lucas Wang, Marianny De León, Boyang Zhou, Sai Jayanth Kalisi, Keith Meng Phou, Jennifer Han, Yijun Chen, Alicia Betty, Oliver Aalami, Alberto Paderno, Rania Hussein. To appear. **CARE - Clinician Augmented Reality Environment: Design and Technical Feasibility of Apple Vision Pro for Image-guided Surgeries**. *Journal of the American Medical Informatics Association (JAMIA)*.

Work experience

Grant and Project Coordinator (20hr/wk) 6/2023 – 6/2025
University of Washington Campus Sustainability Fund, Seattle, WA

- Directed a portfolio of over 50 campus sustainability projects (solar, biodiversity, environmental justice etc.) funded by a \$500,000/year grant.

- Contributed to several campus-as-a-living-lab research projects, focusing on developing tools to solve real-life issues such as mobile app for crowdsourcing bird-building collision data, and data analysis of toxic chemical rainwater runoff.
- Served as the primary liaison between project teams and diverse stakeholders, including tribal representatives, local nonprofits, infrastructure engineers, and research groups. Conducted over 300 meetings.
- Helped implement participatory budgeting and decision-making practices with an eight-member student committee at Campus Sustainability Fund, becoming the first organization at the university to establish such a governance structure.

Software Engineer Intern - GIS (40hr/wk)

6/2024 – 12/2024

King County Department of Natural Resources & Parks, Seattle, WA

- Engineered and deployed new features for Lucity GIS, integrating disparate geospatial (ArcGIS) and asset PostgreSQL databases through gRPC. Streamlined infrastructure maintenance workflows, resulting in a significant reduction in manual data entry for 350+ park maintenance crews, rangers, and environmental scientists. Enabled offline data logging with the ability to upload snapshots and videos.
- Developed Python and ArcGIS-based data analytics pipelines to analyze county demographic and environmental data, identifying five underserved communities and informing updates to the trail master plan for equitable transportation infrastructure development. Presented the trail master plan proposal to executives, which was subsequently adopted into the park levy.

AI Software Engineer Intern (20hr/wk)

6/2023 – 9/2023

Hugsi, Bellevue, WA

- Architected an error-aware RAG pipeline using GPT-3.5 for a Flutter-based iOS app, improving the relevance of personalized gift suggestions on homepage and reducing hallucinations by 65% based on internal evaluations.

Leadership/Teaching experience

Teaching Assistant (CSE557: Computer Graphics & CSE447: NLP)

9/2025 – Present

University of Washington, Seattle, WA

- Developed and extended a C++ codebase for CMU's Scotty3D graphic engine used by 20 master's and Ph.D. students, adding instructional features such as improved test cases, segfault debugging tools, auto-save functionality, and enhanced accessibility of course materials.

Club President

11/2022 – 9/2025

Extended Reality Association at the University of Washington, Seattle, WA

- Led a team of seven officers and over 80 student to build an interdisciplinary Extended Reality community in the Seattle area through hosting technical workshops, speaker series, and hackathons.
- Secured over \$60,000 to purchase Vision Pros and Quest headsets accessible to all students.
- Utilized LiDAR scanning and point cloud processing to digitally reconstruct high-accuracy accessibility maps for two major campus buildings, delivering actionable data to the building management and visitor center.

Talks

"Measuring the impact of student-led sustainability"

2025

Association for the Advancement of Sustainability in Higher Education Conference

"Leading justice-centered sustainability"

2023, 2024

Washington Oregon Higher Education Sustainability Conference

Fellowships

Mary Gates Scholarship

2024

\$5,000

Technical skills

Programming Languages: Java, Python, Rust, C++, Typescript, Swift, C#, GLSL, SQL

Other: Slurm, PyTorch, Google Cloud, AWS, Docker, Apptainer, React, Apache Spark, PostgreSQL, OpenXR, RealityKit, Unity, Unreal, Three.js, ArcGIS, MCP, Prompt Engineering, LlamaIndex